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|                        NUMSYS PRO                        |
|            Multi-Base Number System Calculator           |
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============================================================
|STEP 1: CONVERT FROM                                      |
|Choose a number system:                                   |
============================================================
+------------+  +------------+  +------------+  +------------+
|    [2]     |  |    [8]     |  |    [10]    |  |    [16]    |
|   Binary   |  |   Octal    |  |  Decimal   |  |Hexadecimal |
+------------+  +------------+  +------------+  +------------+
Type 2, 8, 10, or 16 and press Enter: 2

============================================================
|STEP 2: CONVERT TO                                        |
|Choose a number system:                                   |
============================================================
+------------+  +------------+  +------------+  +------------+
|    [2]     |  |    [8]     |  |    [10]    |  |    [16]    |
|   Binary   |  |   Octal    |  |  Decimal   |  |Hexadecimal |
+------------+  +------------+  +------------+  +------------+
Type 2, 8, 10, or 16 and press Enter: 16

============================================================
Enter a binary number
Use only 0 and 1
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Number: 100100111

============================================================
TO DECIMAL (digit x base^position)
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1 x 2^8 = 256
0 x 2^7 = 0
0 x 2^6 = 0
1 x 2^5 = 32
0 x 2^4 = 0
0 x 2^3 = 0
1 x 2^2 = 4
1 x 2^1 = 2
1 x 2^0 = 1
Total = 295

============================================================
FROM DECIMAL (divide by 16 until the quotient is 0)
============================================================
295 / 16 = 18 remainder 7
18 / 16 = 1 remainder 2
1 / 16 = 0 remainder 1
Read the remainders from bottom to top.
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============================================================
RESULT
100100111 (binary)
= 127 (hexadecimal)
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============================================================
WHAT NEXT?
Retry     = same bases, new number
New Bases = choose new bases
Quit      = exit the program
============================================================
+------------+  +------------+  +------------+
|    [R]     |  |    [N]     |  |    [Q]     |
|   Retry    |  | New Bases  |  |    Quit    |
+------------+  +------------+  +------------+
Type R, N, or Q and press Enter: nn

============================================================
Invalid choice!
Please type R, N, or Q.
============================================================

============================================================
WHAT NEXT?
Retry     = same bases, new number
New Bases = choose new bases
Quit      = exit the program
============================================================
+------------+  +------------+  +------------+
|    [R]     |  |    [N]     |  |    [Q]     |
|   Retry    |  | New Bases  |  |    Quit    |
+------------+  +------------+  +------------+
```